

0.1 Bandwidth Selection

$f(x) \rightarrow$ parameter, $y = f(x) + \varepsilon$

$\hat{f}(x) \rightarrow$ estimate, $E[\varepsilon] = 0$

$$\int E \left[f(x) - \hat{f}(x) \right]^2 dx \quad \text{MSE at } x.$$

Y is integrated.

$$\text{MSE} = \text{Bias}^2 + \text{Var}$$

$$\text{bias} = O(h^2), \quad \text{var} = O\left(\frac{1}{nh}\right)$$

$$\text{MSE} = C_1 h^4 + C_2 \frac{1}{nh}$$

minimised at $h \asymp n^{-\frac{1}{5}}$.

Optimal rate: $n^{-4/5}$.

0.1.1 Cross-Validation

fix h

Remark

Leave one out.

Given n points $(x_1, y_1), \dots, (x_n, y_n)$:

1. Take (x_i, y_i) out of the data.
2. Fit Kernel Regression on the rest of the points.
3. Compute

$$\sum_{i=1}^n \left(\hat{f}_{[-i]}(x_i) - y_i \right)^2 \longrightarrow \sum_{i=1}^n e_i^2.$$

This gives $\widehat{\text{MSE}}(h)$. Run this over values of h on a grid. Find h for which $\widehat{\text{MSE}}(h)$ is minimised.

0.1.2 K -Fold Cross-Validation for h

Given n points (x_i, y_i) , $i = 1, \dots, n$.

Procedure: K -Fold Cross-Validation Procedure

1. Divide n data points into K equal parts. Do this N times (e.g. $N = 100$).
2. For each division, repeat for $j = 1, \dots, K$:
 - (a) Take the j -th part (chunk of size $\lfloor n/K \rfloor$). Leave it out. Estimate f based on the remaining $(K - 1)$ parts.
 - (b) Find the error on the j -th part by taking the average of squared errors:

$$\sum_{i \in \{j\}} \left(\hat{f}_{[-j]}(x_i) - y_i \right)^2.$$

3. Sum over $j = 1, \dots, K$.
This yields $\widehat{\text{MSE}}(h)$.

Note

- If $h < O(n^{-1/5})$, then $\frac{1}{nh}$ is of higher order than $n^{-4/5}$.
 - If $h > O(n^{-1/5})$, then h^4 is of higher order than $n^{-4/5}$.
- In either case, the MSE deteriorates; only $h \asymp n^{-1/5}$ achieves the optimal rate $O(n^{-4/5})$.

Procedure: K -Fold Cross-Validation

For h : for one division of n units into K equal parts,

1. **Repeat** for $j = 1, \dots, K$:
 - Take j^{th} part. Leave it out. Estimate f based on the $(K - 1)$ parts.
 - Find the error on the j^{th} part, by taking average of squared errors.
 2. Sum to determine h .
- Repeat N times $\rightarrow \widehat{\text{MSE}}(h)$.